

Written HW5

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I use Neovim (BTW)

Full HW repository at
github.com/LoganCTanner/School

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1. Suppose X and Y have the joint density function $f_{X,Y}(x,y) = e^{-(x+y)}$ for $x \geq 0, y \geq 0$. Let $W = X + Y$.

(a) Are X and Y independent? Justify.

Answer:

If $f_X(x) \cdot f_Y(y) = f_{X,Y}(x,y)$, then they are independent.

$$\begin{aligned} f_X(x) &= e^{-x} \int_0^{\infty} e^{-y} dy \\ &= e^{-x} (-e^{-y}|_0^{\infty}) \\ &= e^{-x} (0 - (-1)) \\ &= e^{-x} \end{aligned}$$

$$\begin{aligned} f_Y(y) &= e^{-y} \int_0^{\infty} e^{-x} dx \\ &= e^{-y} (-e^{-x}|_0^{\infty}) \\ &= e^{-y} (0 - (-1)) \\ &= e^{-y} \end{aligned}$$

$$\begin{aligned} f_X(x) \cdot f_Y(y) &= e^{-x} \cdot e^{-y} \\ &= e^{-(x+y)} \\ &= f_{X,Y}(x,y) \end{aligned}$$

$\therefore f_{X,Y}(x,y)$ is independent.

(b) Find the cdf of W .

Answer:

$$\begin{aligned} F_W(w) &= \int_0^w \int_0^{w-x} e^{-x-y} dy dx \\ &= \int_0^w e^{-x} (-e^{-y}|_0^{w-x}) dx \\ &= \int_0^w e^{-x} (1 - e^{x-w}) dx \\ &= \int_0^w e^{-x} - e^{-x+x-w} dx \\ &= \int_0^w e^{-x} - e^{-w} dx \\ &= -e^{-x} - xe^{-w}|_0^w \\ &= -e^{-w} - we^{-w} - (-1 - 0) \\ &= -e^{-w} - we^{-w} + 1 \end{aligned}$$

(c) Use (b) to find the pdf of W .

Answer:

$$\begin{aligned}f_W(w) &= -e^{-w} - we^{-w} + 1dx \\&= e^{-w} - (e^{-w} + (-we^{-w})) \\&= e^{-w} + we^{-w} - e^{-w} \\&= we^{-w}\end{aligned}$$

(d) Now find the pdf of W using the convolution formula (from the beginning of Chapter 6.3).